

75

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

the PRACTICE PAPER

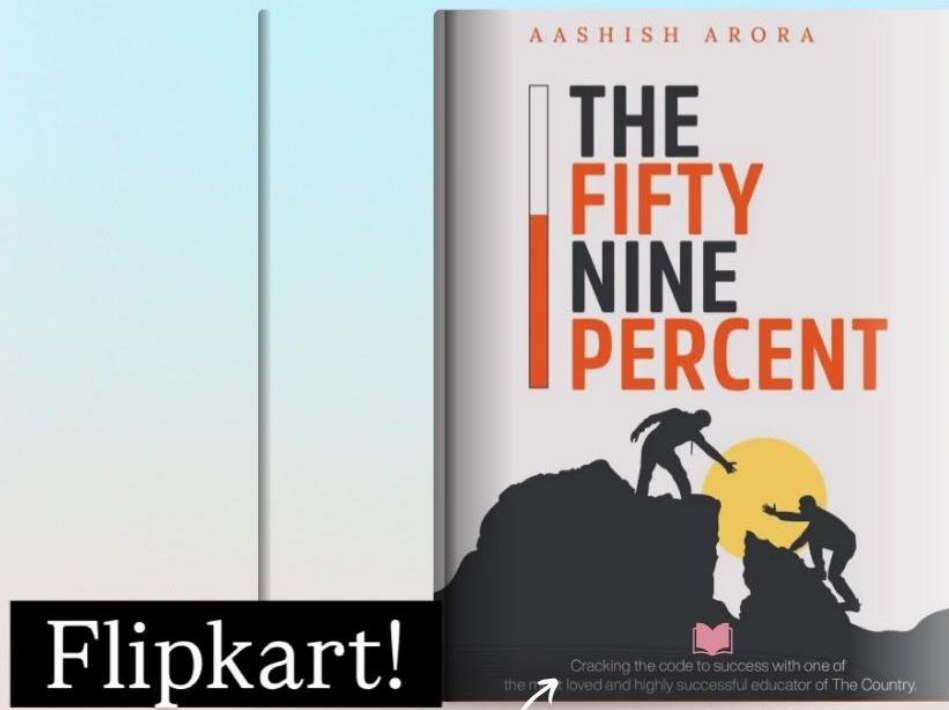


/STUDIFIEDPDF

Subscribe to
STUDIFIEDTM
YouTube Channel and
Learn Quantitative Aptitude
For Bank Exams from India's
Most **Loved** Teacher

THE BOOK YOU LOVED IS BACK

Only at Rs.199/-





COMPLETE QUANT SUBSCRIPTION

FOUNDATION TO ADVANCE BATCH FOR ALL BANK & INSURANCE EXAMS

ENROLL NOW

12 MONTHS	24 MONTHS
₹3,500 ₹1,399	₹4,500 ₹1,799

60% Off



PRELIMS + MAINS

OFFER VALID TILL 5TH JUNE

USE CODE: TT60

 Call or WhatsApp: 08065130831



Extend Your
QUANT
Subscription

12 MONTHS ₹3,500 ₹ 1050	24 MONTHS ₹4,500 ₹ 1350
--	--

USE CODE

→ **RENEWTT**

OFFER VALID TILL 5TH JUNE



Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

CONTENTS

1. SIMPLIFICATION & APPROXIMATION	9
2. ARITHMETIC WORD PROBLEMS	22
3. QUADRATIC EQUATIONS	43
4. WRONG NUMBER SERIES	59
5. MISSING NUMBER SERIES	70
6. DATA INTERPRETATION	82

1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $2\frac{5}{8}$ of 64 + $\frac{1}{7}$ of 98 = 2 ? + $\sqrt{289}$

- A. 78.5
- B. 80.5
- C. 82.5
- D. 84.5
- E. None of these

2. $[(896 + 544) \div 18] + \frac{5}{8}$ of 96 = 3 ? + $\sqrt{25}$

- A. 42
- B. 45
- C. 48
- D. 51
- E. None of these

3. $\frac{11}{12}$ of 288 + 18.75% of 192 = ?² + 11

- A. 14
- B. 15
- C. 16

D. 17

E. None of these

4. $\sqrt[3]{2197} + 1\frac{3}{4}$ of 72 = ?³ + 75

- A. 4
- B. 5
- C. 6
- D. 7
- E. None of these

5. $(15^2 - 9^2) \div 6 + \frac{7}{12}$ of 144 = 4 ? + 6²

- A. 14
- B. 16
- C. 18
- D. 20
- E. None of these

6. 56.25% of 320 - $\frac{1}{8}$ of 256 = ?² + 84

- A. 7
- B. 8
- C. 9
- D. 10

E. None of these

7. $1\frac{5}{8}$ of 80 + $\frac{11}{16}$ of 160 = ? + 44

A. 22

B. 24

C. 26

D. 28

E. None of these

8. $(\sqrt{1024} + \frac{7}{8} \text{ of } 144) \div 2 = ? + 47$

A. 28

B. 30

C. 34

D. 36

E. None of these

9. 37.5% of 240 + 68.75% of 160 = 5 ? + 20

A. 36

B. 38

C. 40

D. 42

E. None of these

10. $(\frac{14}{15} \text{ of } 150 + 32.5\% \text{ of } 200) - \sqrt{625} = ? + 140$

A. 36

B. 38

C. 40

D. 42

E. None of these

11. $[(18^2 + 12^2) \div 18] - \frac{1}{6} \text{ of } 54 + 12.5\% \text{ of } 64 = ?$

A. 21

B. 23

C. 27

D. 29

E. None of these

12. $2\frac{3}{7} \text{ of } 56 + \sqrt[3]{4913} = ?^2 + 9$

A. 10

B. 12

C. 14

D. 16

E. None of these

13. $\frac{9}{16} \text{ of } 320 + \frac{5}{12} \text{ of } 288 = 6 ? + 48$

A. 36

B. 38

C. 40

D. 42

E. None of these

$$14. \sqrt{2025} + \frac{2}{15} \text{ of } 225 + \frac{3}{5} \text{ of } 100 = 5 ?$$

+ 15

A. 24

B. 26

C. 28

D. 30

E. None of these

$$15. (625 \div 25) + \frac{14}{15} \text{ of } 150 = 4 ? + 5^2$$

A. 31

B. 33

C. 37

D. 39

E. None of these

$$16. 1\frac{1}{4} \text{ of } 96 + \frac{8}{15} \text{ of } 150 = 4 ? + 12$$

A. 43

B. 45

C. 47

D. 49

E. None of these

$$17. \sqrt{2401} + \frac{5}{8} \text{ of } 96 - \frac{11}{16} \text{ of } 64 = 2 ? + 5$$

A. 28

B. 30

C. 32

D. 34

E. None of these

$$18. [(11^2 - 5^2) \div 4] + \frac{7}{9} \text{ of } 180 = ?^2 + 43$$

A. 8

B. 9

C. 10

D. 11

E. None of these

$$19. \sqrt[3]{4096} + 37.5\% \text{ of } 160 - \frac{1}{6} \text{ of } 90 = ?^3 + 34$$

A. 2

B. 4

C. 5

D. 6

E. None of these

$$20. (\frac{9}{16} \text{ of } 320 + \frac{7}{9} \text{ of } 180) - \sqrt{169} = 5 ? + 7$$

A. 60

B. 62

C. 64

D. 66

E. None of these

Answers:

1. C
2. B
3. D
4. A
5. C
6. B
7. D
8. E
9. A
10. C
11. E
12. B
13. D
14. A
15. E
16. C
17. B
18. D
19. E
20. A

Solutions:

$$1. 2\frac{5}{8} \text{ of } 64 + \frac{1}{7} \text{ of } 98 = 2 ? + \sqrt{289}$$

$$168 + 14 = 2 ? + 17$$

$$182 = 2 ? + 17$$

$$2 ? = 165$$

$$? = 82.5$$

$$2. [(896 + 544) \div 18] + \frac{5}{8} \text{ of } 96 = 3 ? + \sqrt{25}$$

$$80 + 60 = 3 ? + 5$$

$$140 = 3 ? + 5$$

$$3 ? = 135$$

$$? = 45$$

$$3. \frac{11}{12} \text{ of } 288 + 18.75\% \text{ of } 192 = ?^2 + 11$$

$$264 + 36 = ?^2 + 11$$

$$300 = ?^2 + 11$$

$$?^2 = 289$$

$$? = 17$$

$$4. \sqrt[3]{2197} + 1\frac{3}{4} \text{ of } 72 = ?^3 + 75$$

$$13 + 126 = ?^3 + 75$$

$$139 = ?^3 + 75$$

$$?^3 = 64$$

$$? = 4$$

$$5. (15^2 - 9^2) \div 6 + \frac{7}{12} \text{ of } 144 = 4 ? + 6^2$$

$$(225 - 81) \div 6 + 84 = 4 ? + 36$$

$$144 \div 6 + 84 = 4 ? + 36$$

$$24 + 84 = 4 ? + 36$$

$$108 = 4 ? + 36$$

$$4 ? = 72$$

$$? = 18$$

$$6. 56.25\% \text{ of } 320 - \frac{1}{8} \text{ of } 256 = ?^2 + 84$$

$$180 - 32 = ?^2 + 84$$

$$148 = ?^2 + 84$$

$$?^2 = 64$$

$$? = 8$$

$$7. 1\frac{5}{8} \text{ of } 80 + \frac{11}{16} \text{ of } 160 = 7 ? + 44$$

$$130 + 110 = 7 ? + 44$$

$$240 = 7 ? + 44$$

$$7 ? = 196$$

$$? = 28$$

$$8. (\sqrt{1024} + \frac{7}{8} \text{ of } 144) \div 2 = ? + 47$$

$$(32 + 126) \div 2 = ? + 47$$

$$158 \div 2 = ? + 47$$

$$79 = ? + 47$$

$$? = 32$$

$$9. 37.5\% \text{ of } 240 + 68.75\% \text{ of } 160 = 5 ? + 20$$

$$90 + 110 = 5 ? + 20$$

$$200 = 5 ? + 20$$

$$5 ? = 180$$

$$? = 36$$

$$10. (\frac{14}{15} \text{ of } 150 + 32.5\% \text{ of } 200) - \sqrt{625} = ? + 140$$

$$(140 + 65) - 25 = ? + 140$$

$$205 - 25 = ? + 140$$

$$180 = ? + 140$$

$$? = 40$$

$$11. [(18^2 + 12^2) \div 18] - \frac{1}{6} \text{ of } 54 + 12.5\% \text{ of } 64 = ?$$

$$[(324 + 144) \div 18] - 9 + 8 = ?$$

$$468 \div 18 - 9 + 8 = ?$$

$$26 - 9 + 8 = ?$$

$$? = 25$$

$$12. 2\frac{3}{7} \text{ of } 56 + \sqrt[3]{4913} = ?^2 + 9$$

$$136 + 17 = ?^2 + 9$$

$$153 = ?^2 + 9$$

$$?^2 = 144$$

$$? = 12$$

$$13. \frac{9}{16} \text{ of } 320 + \frac{5}{12} \text{ of } 288 = 6 ? + 48$$

$$180 + 120 = 6 ? + 48$$

$$300 = 6 ? + 48$$

$$6 ? = 252$$

$$? = 42$$

$$14. \sqrt{2025} + \frac{2}{15} \text{ of } 225 + \frac{3}{5} \text{ of } 100 = 5 ? + 15$$

$$45 + 30 + 60 = 5 ? + 15$$

$$135 = 5 ? + 15$$

$$5 ? = 120$$

$$? = 24$$

$$15. (625 \div 25) + \frac{14}{15} \text{ of } 150 = 4 ? + 5^2$$

$$25 + 140 = 4 ? + 25$$

$$165 = 4 ? + 25$$

$$4 ? = 140$$

$$? = 35$$

$$16. 1\frac{1}{4} \text{ of } 96 + \frac{8}{15} \text{ of } 150 = 4 ? + 12$$

$$120 + 80 = 4 ? + 12$$

$$200 = 4 ? + 12$$

$$4 ? = 188$$

$$? = 47$$

$$17. \sqrt{2401} + \frac{5}{8} \text{ of } 96 - \frac{11}{16} \text{ of } 64 = 2 ? + 5$$

$$49 + 60 - 44 = 2 ? + 5$$

$$65 = 2 ? + 5$$

$$2 ? = 60$$

$$? = 30$$

$$18. [(11^2 - 5^2) \div 4] + \frac{7}{9} \text{ of } 180 = ?^2 + 43$$

$$[(121 - 25) \div 4] + 140 = ?^2 + 43$$

$$96 \div 4 + 140 = ?^2 + 43$$

$$24 + 140 = ?^2 + 43$$

$$164 = ?^2 + 43$$

$$?^2 = 121$$

$$? = 11$$

$$19. \sqrt[3]{4096} + 37.5\% \text{ of } 160 - \frac{1}{6} \text{ of } 90 = ?^3 + 34$$

$$16 + 60 - 15 = ?^3 + 34$$

$$61 = ?^3 + 34$$

$$?^3 = 27$$

$$? = 3$$

$$20. (\frac{9}{16} \text{ of } 320 + \frac{7}{9} \text{ of } 180) - \sqrt{169} = 5 ? + 7$$

$$(180 + 140) - 13 = 5 ? + 7$$

$$320 - 13 = 5 ? + 7$$

$$307 = 5 ? + 7$$

$$5 ? = 300$$

$$? = 60$$

2. ARITHMETIC QUESTIONS

Q1: A man increases his speed from 36 km/hr to 48 km/hr and takes 40 minutes less to cover a certain distance D. Find the time taken by him to cover 150% of D at his original speed.

एक व्यक्ति अपनी चाल 36 किमी/घंटा से बढ़ाकर 48 किमी/घंटा कर देता है और एक निश्चित दूरी D तय करने में 40 मिनट कम लेता है। उसकी मूल चाल से D की 150% दूरी तय करने में लगने वाला समय ज्ञात कीजिए।

- (A) 3 hours 20 minutes
- (B) 4 hours
- (C) 4 hours 30 minutes
- (D) 3 hours 45 minutes
- (E) 5 hours

Q2: The sum of two numbers is equal to the sum of the square of 12 and the cube of 8. The larger number is 20 less than the square of 24. Find the sum of twice of 40% of the smaller number and half of the larger number.

दो संख्याओं का योग 12 के वर्ग और 8 के घन के योग के बराबर है। बड़ी संख्या 24 के वर्ग से 20 कम है। छोटी संख्या के 40% के दोगुने और बड़ी संख्या के आधे का योग ज्ञात कीजिए।

- (A) 348
- (B) 368
- (C) 378
- (D) 388
- (E) 358

Q3: The length of a rectangular park is 140% of the radius of a circular park whose circumference is 220 meters. If the area of the rectangular park is 1225 square meters, then the radius of the circular park is what percent more than the breadth of the rectangular park? Take $\pi = 22/7$.

एक आयताकार पार्क की लंबाई एक वृत्ताकार पार्क की त्रिज्या की 140% है, जिसकी परिधि 220 मीटर है। यदि आयताकार पार्क का क्षेत्रफल 1225 वर्ग मीटर है, तो वृत्ताकार पार्क की त्रिज्या आयताकार पार्क की चौड़ाई से कितने प्रतिशत अधिक है? $\pi = 22/7$ लीजिए।

- (A) 40%
- (B) 35%
- (C) 45%

- (D) 50%
- (E) 60%

Q4: A container has 80 ml of a mixture of two liquids A and B. Liquid A is 48 ml more than liquid B. If x ml of liquid A is removed from the container, the ratio of liquid A to liquid B in the remaining mixture becomes 3 : 1. Find x .

एक पात्र में दो द्रव A और B का 80 मिली मिश्रण है। द्रव A, द्रव B से 48 मिली अधिक है। यदि पात्र से द्रव A के x मिली निकाल दिए जाएँ, तो शेष मिश्रण में द्रव A और द्रव B का अनुपात 3 : 1 हो जाता है। x ज्ञात कीजिए।

- (A) 12 ml
- (B) 20 ml
- (C) 24 ml
- (D) 16 ml
- (E) 18 ml

Q5: The ratio of the ages of Person A and Person B after 5 years will be 4 : 7. The ratio of the present ages of Person C and Person B is 3 : 5. If the sum of the ages of Person A and Person C after 3 years will be 80 years, find the sum of the present ages of Person A and Person B.

5 वर्ष बाद व्यक्ति A और व्यक्ति B की आयु का अनुपात 4 : 7 होगा। व्यक्ति C और व्यक्ति B की वर्तमान आयु का अनुपात 3 : 5 है। यदि 3 वर्ष बाद व्यक्ति A और व्यक्ति C की आयु का योग 80 वर्ष होगा, तो व्यक्ति A और व्यक्ति B की वर्तमान आयु का योग ज्ञात कीजिए।

- (A) 88 years
- (B) 96 years
- (C) 100 years
- (D) 108 years
- (E) 92 years

Q6: The curved surface area of a cylinder is 2112 cm^2 and its total surface area is 4884 cm^2 . Find the radius of the cylinder.

एक बेलन का वक्र पृष्ठीय क्षेत्रफल 2112 वर्ग सेमी है और उसका कुल पृष्ठीय क्षेत्रफल 4884 वर्ग सेमी है। बेलन की त्रिज्या ज्ञात कीजिए।

- (A) 18 cm
- (B) 24 cm
- (C) 21 cm
- (D) 28 cm

(E) 14 cm

Q7: There are 36 balls in a bag. Out of these, 8 are red balls, x are yellow balls and the remaining are black balls. If one ball is picked at random, the probability of getting a yellow ball is $\frac{5}{9}$. Find the difference between the number of yellow balls and black balls.

एक थैले में 36 गेंदें हैं। इनमें से 8 लाल गेंदें हैं, x पीली गेंदें हैं और शेष काली गेंदें हैं। यदि एक गेंद यादृच्छिक रूप से निकाली जाती है, तो पीली गेंद आने की प्रायिकता $\frac{5}{9}$ है। पीली गेंदों और काली गेंदों की संख्या का अंतर ज्ञात कीजिए।

- (A) 10
- (B) 8
- (C) 14
- (D) 16
- (E) 12

Q8: A shopkeeper mixed 30 kg of wheat costing Rs 80 per kg with P kg of maize costing Rs 50 per kg. He sold the mixture at Rs 54 per kg and incurred a loss of 10%. Find the value of P .

एक दुकानदार ने Rs 80 प्रति किग्रा वाली 30 किग्रा गेहूं को Rs 50 प्रति किग्रा वाली P किग्रा मक्का के साथ मिलाया। उसने मिश्रण को Rs 54 प्रति किग्रा पर बेचा और उसे 10% की हानि हुई। P का मान ज्ञात कीजिए।

- (A) 45 kg
- (B) 60 kg
- (C) 75 kg
- (D) 90 kg
- (E) 50 kg

Q9: Six years ago, Vikas's age was equal to the average of the present ages of Vikas, Chetan and Pankaj. Pankaj is 10 years older than Chetan. If the present age of Vikas is 45 years, find the present age of Pankaj.

छह वर्ष पहले, विकास की आयु विकास, चेतन और पंकज की वर्तमान आयु के औसत के बराबर थी। पंकज, चेतन से 10 वर्ष बड़ा है। यदि विकास की वर्तमान आयु 45 वर्ष है, तो पंकज की वर्तमान आयु ज्ञात कीजिए।

- (A) 37 years
- (B) 39 years
- (C) 43 years
- (D) 41 years

(E) 45 years

Q10: A woman distributes a sum of money among her daughter, nephew and sister in the ratio of 5:3:2. Her nephew invests his share in a scheme at 25% simple interest per annum for 3 years and earns Rs 900 as interest. Find the share of her sister.

एक महिला अपनी बेटी, भतीजे और बहन के बीच धनराशि को 5:3:2 के अनुपात में बांटती है। उसका भतीजा अपना हिस्सा 3 वर्षों के लिए 25% वार्षिक साधारण ब्याज पर निवेश करता है और Rs 900 ब्याज प्राप्त करता है। उसकी बहन का हिस्सा ज्ञात कीजिए।

- (A) Rs 800
- (B) Rs 1000
- (C) Rs 600
- (D) Rs 1200
- (E) Rs 900

Q11: A milk seller has 120 liters of pure milk. He mixes some water into it and sells the mixture at the cost price of pure milk. If the cost price of pure milk is Rs. 30 per liter and he earns a profit of 20%, find the quantity of water mixed.

एक दूधवाले के पास 120 लीटर शुद्ध दूध है। वह उसमें कुछ पानी मिलाता है और मिश्रण को शुद्ध दूध के क्रय मूल्य पर बेचता है। यदि शुद्ध दूध का क्रय मूल्य Rs. 30 प्रति लीटर है और उसे 20% लाभ होता है, तो मिलाए गए पानी की मात्रा ज्ञात कीजिए।

- (A) 18 liters
- (B) 20 liters
- (C) 24 liters
- (D) 30 liters
- (E) 36 liters

Q12: Five years ago, Aarush's age was 50% more than Bhavin's age and 200% more than Chaitanya's age. Twelve years hence, Chaitanya's age will be 50% of Aarush's age. After $(n + 4)$ years, Bhavin's age will be 50% more than Chaitanya's age. Find n .

पाँच वर्ष पहले, आरुष की आयु भाविन की आयु से 50% अधिक और चैतन्य की आयु से 200% अधिक थी। बारह वर्ष बाद, चैतन्य की आयु आरुष की आयु का 50% होगी। $(n + 4)$ वर्षों बाद, भाविन की आयु चैतन्य की आयु से 50% अधिक होगी। n ज्ञात कीजिए।

- (A) 6
- (B) 8
- (C) 10

(D) 12

(E) 14

Q13: Pipe P alone can fill a tank in 20 minutes, and Pipe Q alone can fill the same tank in X minutes. Pipe R alone can empty the full tank in 60 minutes. If Pipes P, Q and R together fill the tank in 15 minutes, find the time taken by Pipe Q alone to fill $\frac{2}{5}$ of the tank.

पाइप P अकेले एक टंकी को 20 मिनट में भर सकता है, और पाइप Q अकेले उसी टंकी को X मिनट में भर सकता है। पाइप R अकेले पूरी टंकी को 60 मिनट में खाली कर सकता है। यदि पाइप P, Q और R मिलकर टंकी को 15 मिनट में भरते हैं, तो पाइप Q अकेले टंकी का $\frac{2}{5}$ भाग कितने समय में भरेगा?

(A) 9 minutes

(B) 10 minutes

(C) 12 minutes

(D) 15 minutes

(E) 18 minutes

Q14: The average of 12 numbers is 160, and the average of the first 6 numbers is 145. If the average of the last 4 numbers is 210 and the ratio of the seventh number to the eighth number is 4:3, find the eighth number.

12 संख्याओं का औसत 160 है और पहली 6 संख्याओं का औसत 145 है। यदि अंतिम 4 संख्याओं का औसत 210 है और सातवीं संख्या तथा आठवीं संख्या का अनुपात 4:3 है, तो आठवीं संख्या ज्ञात कीजिए।

(A) 72

(B) 84

(C) 90

(D) 96

(E) 108

Q15: The smallest angle of a triangle is equal to $\frac{1}{4}$ of the largest angle of a quadrilateral. The angles of the quadrilateral are in the ratio of 2:3:4:6. The largest angle of the triangle is $\frac{7}{3}$ times its smallest angle. Find the difference between the second largest angle of the triangle and the smallest angle of the quadrilateral.

एक त्रिभुज का सबसे छोटा कोण, एक चतुर्भुज के सबसे बड़े कोण का $\frac{1}{4}$ है। चतुर्भुज के कोणों का अनुपात 2:3:4:6 है। त्रिभुज का सबसे बड़ा कोण उसके सबसे छोटे कोण का $\frac{7}{3}$ गुना है। त्रिभुज के दूसरे सबसे बड़े कोण और चतुर्भुज के सबसे छोटे कोण के बीच का अंतर

ज्ञात कीजिए।

- (A) 12°
- (B) 16°
- (C) 20°
- (D) 24°
- (E) 28°

Q16: $\frac{5}{6}$ of a number is 16 less than $\frac{2}{3}$ of another number. If the sum of both numbers is 240, find the smaller number.

किसी संख्या का $\frac{5}{6}$, दूसरी संख्या के $\frac{2}{3}$ से 16 कम है। यदि दोनों संख्याओं का योग 240 है, तो छोटी संख्या ज्ञात कीजिए।

- (A) 108
- (B) 120
- (C) 96
- (D) 104
- (E) 112

Q17: A person spends 20% of his monthly income on rent and 25% of the remaining income on food. He then spends 40% of the remaining amount on transport and saves Rs. 14400. Find his monthly income.

एक व्यक्ति अपनी मासिक आय का 20% किराए पर और बची हुई आय का 25% भोजन पर खर्च करता है। फिर वह बची हुई राशि का 40% परिवहन पर खर्च करता है और Rs. 14400 बचाता है। उसकी मासिक आय ज्ञात कीजिए।

- (A) Rs. 36000
- (B) Rs. 40000
- (C) Rs. 48000
- (D) Rs. 32000
- (E) Rs. 45000

Q18: A person invested Rs. 1250 at simple interest for X years at X% p.a. and received Rs. 200 as interest. If he invests Rs. 3000 at compound interest for (X - 2) years at (X + 6)% p.a., find the amount received.

एक व्यक्ति ने Rs. 1250 को X वर्षों के लिए X% वार्षिक साधारण ब्याज पर निवेश किया और Rs. 200 ब्याज प्राप्त किया। यदि वह Rs. 3000 को (X - 2) वर्षों के लिए (X + 6)% वार्षिक चक्रवृद्धि ब्याज पर निवेश करता है, तो प्राप्त राशि ज्ञात कीजिए।

- (A) 3440
- (B) 3960

- (C) 3520
- (D) 3630
- (E) 3300

Q19: Six years ago, Aarav's age was 150% more than Nitin's age at that time. Vikas's present age is 150% more than Nitin's present age. If the average of the present ages of Aarav, Nitin and Vikas is equal to Aarav's present age, find Vikas's present age.

छह वर्ष पहले, आरव की आयु उस समय नितिन की आयु से 150% अधिक थी। विकास की वर्तमान आयु, नितिन की वर्तमान आयु से 150% अधिक है। यदि आरव, नितिन और विकास की वर्तमान आयु का औसत आरव की वर्तमान आयु के बराबर है, तो विकास की वर्तमान आयु ज्ञात कीजिए।

- (A) 36 years
- (B) 28 years
- (C) 30 years
- (D) 24 years
- (E) 32 years

Q20: The average weight of 5 students in a group is 42 kg. If one new student joins the group and the average weight decreases by 5 kg, find the weight of the new student.

एक समूह में 5 छात्रों का औसत वजन 42 किग्रा है। यदि एक नया छात्र समूह में शामिल होता है और औसत वजन 5 किग्रा कम हो जाता है, तो नए छात्र का वजन ज्ञात कीजिए।

- (A) 10 kg
- (B) 18 kg
- (C) 20 kg
- (D) 15 kg
- (E) 12 kg

Answers:

- | | |
|------|-------|
| 1. B | 8. B |
| 2. E | 9. D |
| 3. A | 10. A |
| 4. D | 11. C |
| 5. C | 12. B |
| 6. C | 13. C |
| 7. E | 14. C |

15.A
16.C
17.B

18.D
19.C
20.E

Solutions:

1. Option B

Original speed = 36 km/hr

Increased speed = 48 km/hr

Time saved = 40 minutes = $40/60 = 2/3$ hour

Let the distance be D km.

Time taken at original speed = $D/36$

Time taken at increased speed = $D/48$

According to the question:

$$D/36 - D/48 = 2/3$$

$$\text{LCM of 36 and 48} = 144$$

$$4D/144 - 3D/144 = 2/3$$

$$D/144 = 2/3$$

$$D = 144 \times 2/3$$

$$D = 96 \text{ km}$$

$$\text{Now, } 150\% \text{ of } D = 96 \times 150/100 = 144 \text{ km}$$

$$\text{Time taken at original speed} = 144/36 = 4 \text{ hours}$$

2. Option E

Sum of two numbers = Square of 12 + Cube of 8

$$\text{Square of 12} = 12 \times 12 = 144$$

$$\text{Cube of 8} = 8 \times 8 \times 8 = 512$$

$$\text{Sum of two numbers} = 144 + 512 = 656$$

$$\text{Larger number} = \text{Square of 24} - 20$$

$$\text{Larger number} = 24 \times 24 - 20 = 576 - 20 = 556$$

$$\text{Smaller number} = 656 - 556 = 100$$

$$40\% \text{ of smaller number} = 100 \times 40/100 = 40$$

$$\text{Twice of } 40\% \text{ of smaller number} = 2 \times 40 = 80$$

$$\text{Half of larger number} = 556/2 = 278$$

$$\text{Required sum} = 80 + 278 = 358$$

3. Option A

Circumference of circular park = 220 meters

Formula of circumference = $2 \times \pi \times \text{radius}$

Using $\pi = 22/7$:

$$\text{Radius} = 220 \times 7 / (2 \times 22)$$

$$\text{Radius} = 220 \times 7 / 44$$

$$\text{Radius} = 5 \times 7 = 35 \text{ meters}$$

Length of rectangular park = 140% of radius

$$\text{Length} = 35 \times 140 / 100 = 49 \text{ meters}$$

Area of rectangular park = 1225 square meters

Area of rectangle = Length $\times$ Breadth

$$\text{Breadth} = \text{Area} / \text{Length}$$

$$\text{Breadth} = 1225 / 49 = 25 \text{ meters}$$

$$\text{Radius} = 35 \text{ meters}$$

$$\text{Breadth} = 25 \text{ meters}$$

$$\text{Difference} = 35 - 25 = 10 \text{ meters}$$

$$\text{Required percentage more} = \text{Difference} \times 100 / \text{Breadth}$$

$$\text{Required percentage more} = 10 \times 100 / 25 = 40\%$$

4. Option D

Total mixture = 80 ml

Let quantity of liquid B be y ml.

Liquid A is 48 ml more than liquid B.

So, quantity of liquid A = $y + 48$

Now,

$$y + y + 48 = 80$$

$$2y + 48 = 80$$

$$2y = 80 - 48$$

$$2y = 32$$

$$y = 32 / 2 = 16$$

Quantity of liquid B = 16 ml

Quantity of liquid A = $16 + 48 = 64$ ml

Now, x ml of liquid A is removed.

Remaining quantity of liquid A = $64 - x$

Quantity of liquid B remains = 16 ml

New ratio of liquid A to liquid B = 3 : 1

$$\text{So, } (64 - x)/16 = 3/1$$

$$64 - x = 48$$

$$x = 64 - 48 = 16$$

5. Option C

After 5 years, the ratio of ages of Person A and Person B will be 4 : 7.

Let their ages after 5 years be $4x$ and $7x$.

$$\text{Present age of Person A} = 4x - 5$$

$$\text{Present age of Person B} = 7x - 5$$

$$\text{Present age ratio of Person C and Person B} = 3 : 5$$

So, present age of Person C = $3/5$ of Person B

$$\text{Present age of Person C} = 3/5 \times (7x - 5)$$

After 3 years, sum of ages of Person A and Person C = 80 years.

$$\text{So, present sum of their ages} = 80 - 6 = 74 \text{ years}$$

Now,

$$(4x - 5) + 3/5 \times (7x - 5) = 74$$

Multiply by 5:

$$5(4x - 5) + 3(7x - 5) = 370$$

$$20x - 25 + 21x - 15 = 370$$

$$41x - 40 = 370$$

$$41x = 410$$

$$x = 410/41 = 10$$

$$\text{Present age of Person A} = 4 \times 10 - 5 = 35 \text{ years}$$

$$\text{Present age of Person B} = 7 \times 10 - 5 = 65 \text{ years}$$

$$\text{Required sum} = 35 + 65 = 100 \text{ years}$$

6. Option C

Curved surface area of cylinder = 2112 cm^2 .

Total surface area of cylinder = 4884 cm^2 .

In a cylinder:

Total surface area – Curved surface area = area of two circular bases.

$$\text{So, } 2\pi r^2 = 4884 - 2112$$

$$2\pi r^2 = 2772$$

Use $\pi = 22/7$.

$$2 \times 22/7 \times r^2 = 2772$$

$$44r^2/7 = 2772$$

$$r^2 = 2772 \times 7/44$$

$$r^2 = 441$$

$$r = 21 \text{ cm}$$

7. Option E

Total balls in the bag = 36.

Red balls = 8.

Yellow balls = x .

Probability of getting a yellow ball = $5/9$.

$$\text{So, } x/36 = 5/9$$

$$x = 36 \times 5/9$$

$$x = 20$$

Yellow balls = 20.

Black balls = Total balls – Red balls – Yellow balls.

$$\text{Black balls} = 36 - 8 - 20$$

$$\text{Black balls} = 8$$

$$\text{Required difference} = 20 - 8$$

$$\text{Required difference} = 12$$

8. Option B

Selling price of the mixture = Rs 54 per kg.

Loss = 10%.

So, selling price = 90% of cost price.

$$\text{Average cost price of mixture} = 54 \times 100/90$$

$$\text{Average cost price of mixture} = \text{Rs } 60 \text{ per kg.}$$

Now use alligation:

Cost price of wheat = Rs 80 per kg.

Cost price of maize = Rs 50 per kg.

Mean cost price = Rs 60 per kg.

$$\text{Ratio of wheat to maize} = (60 - 50) : (80 - 60)$$

$$\text{Ratio} = 10 : 20$$

$$\text{Ratio} = 1 : 2$$

Wheat quantity = 30 kg = 1 part.

So, maize quantity = 2 parts.

$$\text{Maize quantity} = 2 \times 30 = 60 \text{ kg.}$$

9. Option D

Present age of Vikas = 45 years.

Age of Vikas six years ago = $45 - 6 = 39$ years.

Let the present age of Chetan be x years.

Then present age of Pankaj = $x + 10$ years.

According to the question:

Average of present ages of Vikas, Chetan and Pankaj = 39

So, $(45 + x + x + 10)/3 = 39$

$(55 + 2x)/3 = 39$

$55 + 2x = 39 \times 3$

$55 + 2x = 117$

$2x = 117 - 55$

$2x = 62$

$x = 31$

Present age of Pankaj = $x + 10$

Present age of Pankaj = $31 + 10$

Present age of Pankaj = 41 years

10. Option A

Ratio of shares of daughter, nephew and sister = 5:3:2.

Nephew invests his share at 25% simple interest per annum for 3 years.

Interest earned by nephew = Rs 900.

Simple interest = $\text{Principal} \times \text{Rate} \times \text{Time}/100$

So, $900 = \text{Principal} \times 25 \times 3/100$

$900 = \text{Principal} \times 75/100$

$\text{Principal} = 900 \times 100/75$

$\text{Principal} = \text{Rs } 1200$

So, nephew's share = Rs 1200.

In the ratio, nephew's share = 3 parts.

So, 3 parts = Rs 1200.

1 part = $1200/3 = \text{Rs } 400$.

Sister's share = 2 parts.

Sister's share = $2 \times 400 = \text{Rs } 800$.

11. Option C

Quantity of pure milk = 120 liters

Cost price of pure milk = Rs. 30 per liter

Total cost price = $120 \times 30 = \text{Rs. } 3600$

Let the quantity of water mixed be x liters.

Total mixture = $120 + x$ liters

The mixture is sold at Rs. 30 per liter.

Selling price = $(120 + x) \times 30$

Profit = 20% of Rs. 3600

Profit = $20 \times 3600/100 = \text{Rs. } 720$

So, selling price = $3600 + 720 = \text{Rs. } 4320$

Now:

$(120 + x) \times 30 = 4320$

$120 + x = 4320/30$

$120 + x = 144$

$x = 144 - 120$

$x = 24$

12. Option B

Five years ago, Aarush's age was 50% more than Bhavin's age.

So, Aarush : Bhavin = $150 : 100 = 3 : 2$

Five years ago, Aarush's age was 200% more than Chaitanya's age.

So, Aarush : Chaitanya = $300 : 100 = 3 : 1$

Therefore, five years ago:

Aarush : Bhavin : Chaitanya = $3 : 2 : 1$

Let their ages five years ago be $3x$, $2x$ and x respectively.

Twelve years hence means 17 years after the ages taken five years ago.

Aarush's age after 12 years from present = $3x + 17$

Chaitanya's age after 12 years from present = $x + 17$

Given, Chaitanya's age will be 50% of Aarush's age.

So:

$x + 17 = 50 \times (3x + 17)/100$

$x + 17 = (3x + 17)/2$

$2x + 34 = 3x + 17$

$x = 17$

Present age of Bhavin = $2x + 5 = 34 + 5 = 39$ years

Present age of Chaitanya = $x + 5 = 17 + 5 = 22$ years

After $(n + 4)$ years:

Bhavin's age = $39 + n + 4 = n + 43$

Chaitanya's age = $22 + n + 4 = n + 26$

Given, Bhavin's age will be 50% more than Chaitanya's age.

So:

$$n + 43 = 150 \times (n + 26)/100$$

$$n + 43 = 3(n + 26)/2$$

$$2n + 86 = 3n + 78$$

$$n = 8$$

13. Option C

Pipe P fills the tank in 20 minutes.

Pipe P's 1-minute work = $1/20$

Pipe Q fills the tank in X minutes.

Pipe Q's 1-minute work = $1/X$

Pipe R empties the tank in 60 minutes.

Pipe R's 1-minute work = $-1/60$

Together, they fill the tank in 15 minutes.

So, their 1-minute work = $1/15$

Now:

$$1/20 + 1/X - 1/60 = 1/15$$

$$1/X = 1/15 - 1/20 + 1/60$$

$$\text{LCM of 15, 20 and 60} = 60$$

$$1/X = 4/60 - 3/60 + 1/60$$

$$1/X = 2/60$$

$$1/X = 1/30$$

$$X = 30$$

Pipe Q alone fills the full tank in 30 minutes.

Time taken by Pipe Q to fill $2/5$ of the tank = $30 \times 2/5$

Time taken = 12 minutes

14. Option C

Average of 12 numbers = 160

Sum of 12 numbers = $12 \times 160 = 1920$

Average of first 6 numbers = 145

Sum of first 6 numbers = $6 \times 145 = 870$

Average of last 4 numbers = 210

Sum of last 4 numbers = $4 \times 210 = 840$

So, sum of seventh and eighth numbers = $1920 - 870 - 840$

Sum of seventh and eighth numbers = 210

Ratio of seventh number to eighth number = 4:3

Total ratio parts = $4 + 3 = 7$

$$\text{Eighth number} = 210 \times \frac{3}{7}$$

$$\text{Eighth number} = 90$$

15. Option A

Angles of the quadrilateral are in the ratio 2:3:4:6.

$$\text{Sum of angles of a quadrilateral} = 360^\circ$$

$$\text{Total ratio parts} = 2 + 3 + 4 + 6 = 15$$

$$\text{Smallest angle of quadrilateral} = 360 \times \frac{2}{15} = 48^\circ$$

$$\text{Largest angle of quadrilateral} = 360 \times \frac{6}{15} = 144^\circ$$

$$\text{Smallest angle of triangle} = \frac{1}{4} \text{ of largest angle of quadrilateral}$$

$$\text{Smallest angle of triangle} = \frac{144}{4} = 36^\circ$$

$$\text{Largest angle of triangle} = \frac{7}{3} \times 36$$

$$\text{Largest angle of triangle} = 84^\circ$$

$$\text{Sum of angles of a triangle} = 180^\circ$$

$$\text{Second largest angle of triangle} = 180 - 36 - 84$$

$$\text{Second largest angle of triangle} = 60^\circ$$

$$\text{Required difference} = 60^\circ - 48^\circ$$

$$\text{Required difference} = 12^\circ$$

16. Option C

Let the first number be x.

$$\text{Then the second number} = 240 - x.$$

According to the question:

$$\frac{5}{6} \text{ of first number} = \frac{2}{3} \text{ of second number} - 16.$$

So,

$$\frac{5x}{6} = \frac{2(240 - x)}{3} - 16$$

Multiply the whole equation by 6:

$$5x = 4(240 - x) - 96$$

$$5x = 960 - 4x - 96$$

$$5x = 864 - 4x$$

$$5x + 4x = 864$$

$$9x = 864$$

$$x = \frac{864}{9}$$

$$x = 96$$

$$\text{Second number} = 240 - 96 = 144.$$

$$\text{Smaller number} = 96.$$

17. Option B

Let the monthly income be Rs. x .

Rent = 20% of x .

Remaining income after rent = 80% of x .

Food expense = 25% of remaining income.

Remaining income after food = 75% of 80% of x .

Remaining income after food = $75/100 \times 80/100 \times x$

Remaining income after food = $60x/100$.

Transport expense = 40% of this remaining amount.

So, saving = 60% of this remaining amount.

Saving = $60/100 \times 60x/100$

Saving = $36x/100$.

Given saving = Rs. 14400.

So,

$36x/100 = 14400$

$x = 14400 \times 100/36$

$x = 40000$

18. Option D

Simple interest = Principal $\times$ Rate $\times$ Time/100.

Here, principal = Rs. 1250, rate = $X\%$ and time = X years.

Interest received = Rs. 200.

So,

$1250 \times X \times X/100 = 200$

$1250X^2/100 = 200$

$25X^2/2 = 200$

$25X^2 = 400$

$X^2 = 400/25$

$X^2 = 16$

$X = 4$

Now, Rs. 3000 is invested at compound interest.

Time = $X - 2 = 4 - 2 = 2$ years.

Rate = $X + 6 = 4 + 6 = 10\%$ p.a.

Amount = $3000 \times 110/100 \times 110/100$

Amount = $3000 \times 121/100$

Amount = 3630

19. Option C

Let Nitin's present age be x years.

Six years ago, Nitin's age = $x - 6$.

Aarav's age six years ago was 150% more than Nitin's age at that time.

150% more means 250% of Nitin's age.

So, Aarav's age six years ago = $250/100 \times (x - 6)$

Aarav's age six years ago = $5/2 \times (x - 6)$.

Aarav's present age = $5/2 \times (x - 6) + 6$

Aarav's present age = $5x/2 - 15 + 6$

Aarav's present age = $5x/2 - 9$.

Vikas's present age is 150% more than Nitin's present age.

So, Vikas's present age = 250% of x .

Vikas's present age = $5x/2$.

Average of present ages of all three is equal to Aarav's present age.

So,

Aarav + Nitin + Vikas = $3 \times$ Aarav

Nitin + Vikas = $2 \times$ Aarav

$x + 5x/2 = 2(5x/2 - 9)$

$7x/2 = 5x - 18$

$7x = 10x - 36$

$10x - 7x = 36$

$3x = 36$

$x = 36/3$

$x = 12$

Vikas's present age = $5x/2$.

Vikas's present age = $5 \times 12/2 = 30$ years.

20. Option E

Initial number of students = 5.

Initial average weight = 42 kg.

Initial total weight = $42 \times 5 = 210$ kg.

After one new student joins, total number of students = 6.

Average weight decreases by 5 kg.

New average weight = $42 - 5 = 37$ kg.

New total weight = $37 \times 6 = 222$ kg.

Weight of new student = New total weight - Initial total weight.

Weight of new student = $222 - 210 = 12$ kg.

3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $5x^2 - 51x + 54 = 0$

II. $4y^2 - y - 3 = 0$

2.) I. $6x^2 + x - 2 = 0$

II. $5y^2 - 19y + 12 = 0$

3.) I. $4x^2 - 25x + 36 = 0$

II. $12y^2 - 23y - 9 = 0$

4.) I. $4x^2 - 3x - 10 = 0$

II. $3y^2 - 27y + 42 = 0$

5.) I. $x^2 - 9\sqrt{5}x + 100 = 0$

II. $y^2 - 3\sqrt{5}y + 10 = 0$

6.) I. $2x^2 - 19x + 24 = 0$

II. $7y^2 - 56y + 84 = 0$

7.) I. $2x^2 + 5x - 3 = 0$

II. $4y^2 - 13y + 10 = 0$

8.) I. $3x^2 - 17x + 20 = 0$

II. $6y^2 - 7y - 5 = 0$

9.) I. $8x^2 - 72x + 144 = 0$

II. $10y^2 - 17y - 20 = 0$

10.) I. $9x^2 - 9x - 54 = 0$

II. $7y^2 - 49y + 84 = 0$

11.) I. $10x^2 - x - 2 = 0$

II. $2y^2 - 11y + 12 = 0$

12.) I. $x^2 - 5\sqrt{5}x + 30 = 0$

II. $y^2 - 7\sqrt{5}y + 60 = 0$

13.) I. $11x^2 - 55x - 66 = 0$

II. $5y^2 - 30y + 40 = 0$

14.) I. $3x^2 - 25x + 42 = 0$

II. $9y^2 - 15y - 14 = 0$

15.) I. $2x^2 - 21x + 52 = 0$

II. $5y^2 - 9y - 2 = 0$

16.) I. $6x^2 - 19x - 20 = 0$

II. $3y^2 - 29y + 70 = 0$

17.) I. $5x^2 - 6x - 11 = 0$

II. $10y^2 - 67y + 99 = 0$

18.) I. $x^2 - 7\sqrt{2}x + 20 = 0$

II. $y^2 - 7\sqrt{2}y + 24 = 0$

19.) I. $7x^2 - 63x + 140 = 0$

II. $3y^2 - 19y + 28 = 0$

20.) I. $3x^2 - 35x + 72 = 0$

II. $4y^2 - 27y + 35 = 0$

Answers:

1. A

2. B

3. D

4. E

5. A

6. C

7. B

8. D

9. A

10. E

11. B

12. E

13. C

14. D

15. A

16. B

17. E

18. C

19. D

20. C

Solutions:

$$(1) x = 9, \frac{6}{5}$$

$$y = 1, \frac{3}{4}$$

$$(2) x = \frac{1}{2}, \frac{2}{3}$$

$$y = 3, \frac{4}{5}$$

$$(3) x = 4, \frac{9}{4}$$

$$y = \frac{9}{4}, \frac{1}{3}$$

$$(4) x = 2, \frac{5}{4}$$

$$y = 7, 2$$

$$(5) x = 5\sqrt{5}, 4\sqrt{5}$$

$$y = 2\sqrt{5}, 1\sqrt{5}$$

$$(6) x = 8, \frac{3}{2}$$

$$y = 6, 2$$

$$(7) x = \frac{1}{2}, -3$$

$$y = 2, \frac{5}{4}$$

$$(8) x = 4, \frac{5}{3}$$

$$y = \frac{5}{3}, -\frac{1}{2}$$

$$(9) x = 6, 3$$

$$y = \frac{5}{2}, -\frac{4}{5}$$

$$(10) x = 3, -2$$

$$y = 4, 3$$

$$(11) x = \frac{1}{2}, \frac{2}{5}$$

$$y = 4, \frac{3}{2}$$

$$(12) x = 3\sqrt{5}, 2\sqrt{5}$$

$$y = 4\sqrt{5}, 3\sqrt{5}$$

$$(13) x = 6, -1$$

$$y = 4, 2$$

$$(14) x = 6, \frac{7}{3}$$

$$y = \frac{7}{3}, -\frac{2}{3}$$

$$(15) x = \frac{13}{2}, 4$$

$$y = 2, \frac{1}{5}$$

$$(16) x = 4, \frac{5}{6}$$

$$y = 5, \frac{14}{3}$$

$$(17) x = \frac{11}{5}, -1$$

$$y = \frac{9}{2}, \frac{11}{5}$$

$$(18) x = 5\sqrt{2}, 2\sqrt{2}$$

$$y = 4\sqrt{2}, 3\sqrt{2}$$

$$(19) x = 5, 4$$

$$y = 4, \frac{7}{3}$$

$$(20) x = 9, \frac{8}{3}$$

$$y = 5, \frac{7}{4}$$

4. WRONG NUMBER SERIES

(1) 1845, 1872, 1911, 1960, 2037, 2130

- (a) 1960
- (b) 1872
- (c) 2037
- (d) 2130
- (e) None of these

(2) 4720, 4686, 4634, 4558, 4438, 4310

- (a) 4558
- (b) 4438
- (c) 4686
- (d) 4310
- (e) None of these

(3) 94, 269, 787, 2304, 6860, 20515

- (a) 269
- (b) 6860
- (c) 787
- (d) 2304
- (e) None of these

(4) 6144, 3086, 1564, 798, 440, 262

- (a) 1564
- (b) 440
- (c) 3086
- (d) 798
- (e) None of these

(5) 420, 436, 472, 528, 636, 780

- (a) 436
- (b) 472
- (c) 636
- (d) 780
- (e) None of these

(6) 146, 293, 881, 3527, 17639, 105850

- (a) 105850
- (b) 17639
- (c) 293
- (d) 3527
- (e) None of these

(7) 1200, 1194, 1176, 1128, 1044, 858

- (a) 1176
- (b) 1128

- | | |
|---|--|
| (c) 1044 | (a) 610 |
| (d) 1194 | (b) 512 |
| (e) None of these | (c) 532 |
| | (d) 700 |
| (8) 980, 1000, 1030, 1085, 1150, 1260 | (e) None of these |
| (a) 1030 | |
| (b) 1150 | (12) 145, 282, 830, 3298, 16376, 98128 |
| (c) 1085 | (a) 830 |
| (d) 1260 | (b) 3298 |
| (e) None of these | (c) 282 |
| | (d) 98128 |
| (9) 620, 635, 1255, 1890, 3150, 5035 | (e) None of these |
| (a) 1255 | |
| (b) 635 | (13) 100, 180, 294, 452, 648, 900 |
| (c) 5035 | (a) 648 |
| (d) 3150 | (b) 180 |
| (e) None of these | (c) 452 |
| | (d) 294 |
| (10) 5000, 4488, 4232, 4104, 4055, 4008 | (e) None of these |
| (a) 4488 | |
| (b) 4232 | (14) 15, 112, 678, 3390, 13584, 40755 |
| (c) 4104 | (a) 678 |
| (d) 4008 | (b) 112 |
| (e) None of these | (c) 13584 |
| | (d) 3390 |
| (11) 500, 512, 532, 564, 610, 700 | (e) None of these |

(15) 210, 425, 860, 1725, 3490, 7005

(a) 425

(b) 860

(c) 3490

(d) 7005

(e) None of these

(16) 850, 865, 887, 916, 958, 995

(a) 958

(b) 887

(c) 916

(d) 995

(e) None of these

(17) 220, 433, 852, 1686, 3296, 6480

(a) 852

(b) 1686

(c) 3296

(d) 433

(e) None of these

(18) 1800, 1780, 1758, 1711, 1658, 1590

(a) 1780

(b) 1711

(c) 1758

(d) 1590

(e) None of these

(19) 720, 765, 745, 835, 795, 960

(a) 765

(b) 835

(c) 745

(d) 960

(e) None of these

(20) 1380, 1406, 1449, 1511, 1594, 1712

(a) 1406

(b) 1449

(c) 1511

(d) 1594

(e) None of these

Answers:

(1) a

(2) b

(3) c

(4) d

(5) e

(6) a

(7) b

(8) c

(9) d

(10) e

(11) a

(12) b

(13) c

(14) d

(15) e

(16) a

(17) b

(18) c

(19) d

(20) e

Solutions:

(1) +27, +39, +54, +72, +93; second differences: +12, +15, +18, +21.

(2) -34, -52, -76, -106, -142; second differences: -18, -24, -30, -36.

(3) $\times 3 - 13$, $\times 3 - 26$, $\times 3 - 39$, $\times 3 - 52$, $\times 3 - 65$.

(4) $\div 2 + 14$, $\div 2 + 21$, $\div 2 + 28$, $\div 2 + 35$, $\div 2 + 42$.

(5) $+4^2$, $+6^2$, $+8^2$, $+10^2$, $+12^2$.

(6) $\times 2 + 1$, $\times 3 + 2$, $\times 4 + 3$, $\times 5 + 4$, $\times 6 + 5$.

(7) -6, -18, -42, -90, -186; each next difference = previous difference $\times 2 + 6$.

(8) $+(2 \times 10)$, $+(3 \times 10)$, $+(5 \times 10)$, $+(7 \times 10)$, $+(11 \times 10)$.

(9) Sum of the previous two numbers.

(10) -512, -256, -128, -64, -32.

(11) +12, +20, +32, +52, +84; each difference = sum of previous two differences.

(12) $\times 2 - 8$, $\times 3 - 16$, $\times 4 - 32$, $\times 5 - 64$, $\times 6 - 128$.

(13) $5^3 - 5^2$, $6^3 - 6^2$, $7^3 - 7^2$, $8^3 - 8^2$, $9^3 - 9^2$, $10^3 - 10^2$.

(14) $\times 7 + 7$, $\times 6 + 6$, $\times 5 + 5$, $\times 4 + 4$, $\times 3 + 3$.

(15) $\times 2 + 5$, $\times 2 + 10$, $\times 2 + 15$, $\times 2 + 20$, $\times 2 + 25$.

(16) +15, +22, +29, +36, +43; second differences: +7, +7, +7, +7.

(17) $\times 2 - 7$, $\times 2 - 14$, $\times 2 - 28$, $\times 2 - 56$, $\times 2 - 112$.

(18) -20, -29, -40, -53, -68; second differences: -9, -11, -13, -15.

(19) +45, -20, +90, -40, +180.

(20) +26, +43, +62, +83, +106; second differences: +17, +19, +21, +23.

5. MISSING NUMBER SERIES

(1) 5400, 5388, ?, 5316, 5220, 5028

- (a) 5358
- (b) 5362
- (c) 5364
- (d) 5368
- (e) None of these

(2) 5, 13, 31, 69, ?, 305

- (a) 141
- (b) 143
- (c) 145
- (d) 147
- (e) None of these

(3) 24, 33, 47, ?, 98, 139

- (a) 66
- (b) 68
- (c) 70
- (d) 72
- (e) None of these

(4) 118, ?, 332, 546, 885, 1438

- (a) 207
- (b) 209
- (c) 211
- (d) 213
- (e) None of these

(5) 11520, 2880, 960, ?, 120, 40

- (a) 460
- (b) 470
- (c) 480

(d) 490

(e) None of these

(6) 51.8, 56, ?, 70.7, 81.2, 93.8

- (a) 62.1
- (b) 62.3
- (c) 62.5
- (d) 62.7

(e) None of these

(7) 960, 956, 940, ?, 840, 740

- (a) 896
- (b) 900
- (c) 902
- (d) 904
- (e) None of these

(8) 31, 39, 66, ?, 255, 471

- (a) 126
- (b) 128
- (c) 130
- (d) 132
- (e) None of these

(9) 38, 113, 336, ?, 3002, 8997

- (a) 999
- (b) 1001
- (c) 1003
- (d) 1005
- (e) None of these

(10) 4, 9, ?, 28, 49, 85

- (a) 15
(b) 17
(c) 19
(d) 21
(e) None of these
- (11) 2020, 2037, 2056, ?, 2108, 2139
- (a) 2073
(b) 2075
(c) 2079
(d) 2083
(e) None of these
- (12) 176, 194, ?, 271, 334, 416
- (a) 223
(b) 225
(c) 227
(d) 229
(e) None of these
- (13) 5680, 2850, 960, ?, 60, 20
- (a) 240
(b) 245
(c) 250
(d) 255
(e) None of these
- (14) 36, 56, ?, 128, 184, 256
- (a) 80
(b) 82
(c) 84
(d) 86
(e) None of these
- (15) 86, 176, 361, ?, 1501, 3038
- (a) 738
(b) 742
(c) 746
(d) 750
(e) None of these

- (16) 4320, ?, 4281, 4233, 4158, 4050
- (a) 4304
(b) 4308
(c) 4312
(d) 4316
(e) None of these
- (17) 5, 19, 74, 293, ?, 4667
- (a) 1158
(b) 1162
(c) 1164
(d) 1168
(e) None of these
- (18) 64, 74, 99, ?, 234, 364
- (a) 145
(b) 147
(c) 149
(d) 151
(e) None of these
- (19) 1032, 522, ?, 142, 80, 50
- (a) 264
(b) 268
(c) 272
(d) 276
(e) None of these
- (20) 74, 102, 158, ?, 494, 942
- (a) 270
(b) 272
(c) 274
(d) 276
(e) None of these

Answers:

- (1) c
(2) d

- (3) b
 (4) a
 (5) c
 (6) b
 (7) d
 (8) c
 (9) c
 (10) a
 (11) c
 (12) b
 (13) c
 (14) d
 (15) a
 (16) b
 (17) d
 (18) c
 (19) b
 (20) a

Solutions:

- (1) -12, -24, -48, -96, -192
 (2) $\times 2 + 3$, $\times 2 + 5$, $\times 2 + 7$, $\times 2 + 9$, $\times 2 + 11$
 (3) +9, +14, +21, +30, +41
 Pattern: $+(2^2 + 5)$, $+(3^2 + 5)$, $+(4^2 + 5)$, $+(5^2 + 5)$, $+(6^2 + 5)$
 (4) Sum of previous two numbers + 7
 (5) $\div 4$, $\div 3$, $\div 2$, $\div 4$, $\div 3$
 (6) +4.2, +6.3, +8.4, +10.5, +12.6

$$(7) -2^2, -4^2, -6^2, -8^2, -10^2$$

$$(8) +2^3, +3^3, +4^3, +5^3, +6^3$$

$$(9) \times 3 - 1, \times 3 - 3, \times 3 - 5, \times 3 - 7, \times 3 - 9$$

$$(10) \text{ Sum of previous two numbers } +2, +4, +6, +8$$

$$(11) +17, +19, +23, +29, +31$$

$$(12) +18, +31, +46, +63, +82$$

$$\text{Second difference: } +13, +15, +17, +19$$

$$(13) \div 2 + 10, \div 3 + 10, \div 4 + 10, \div 5 + 10, \div 6 + 10$$

$$(14) +4 \times 5, +5 \times 6, +6 \times 7, +7 \times 8, +8 \times 9$$

$$(15) \times 2 + 2^2, \times 2 + 3^2, \times 2 + 4^2, \times 2 + 5^2, \times 2 + 6^2$$

$$(16) -3 \times 2^2, -3 \times 3^2, -3 \times 4^2, -3 \times 5^2, -3 \times 6^2$$

$$(17) \times 4 - 1, \times 4 - 2, \times 4 - 3, \times 4 - 4, \times 4 - 5$$

$$(18) +10, +25, +50, +85, +130$$

$$\text{Second difference: } +15, +25, +35, +45$$

$$(19) \div 2 + 6, \div 2 + 7, \div 2 + 8, \div 2 + 9, \div 2 + 10$$

$$(20) +28, +56, +112, +224, +448$$

6. DATA INTERPRETATION

Set 1: The table shows the difference between male and female members in five different yoga studios. It also shows the ratio of female members to male members in these studios. Read the data carefully and answer the questions.

Note: The difference is the absolute difference between male and female members.

तालिका पाँच अलग-अलग योग स्टूडियो में पुरुष और महिला सदस्यों के बीच का अंतर दर्शाती है। यह इन स्टूडियो में महिला सदस्यों और पुरुष सदस्यों का अनुपात भी दर्शाती है। आँकड़ों को ध्यानपूर्वक पढ़िए और प्रश्नों के उत्तर दीजिए। नोट: अंतर पुरुष और महिला सदस्यों के बीच का वास्तविक अंतर है।

Yoga Studio / योग स्टूडियो	Difference between male and female members / पुरुष और महिला सदस्यों के बीच का अंतर	Ratio of female to male members / महिला और पुरुष सदस्यों का अनुपात
Kshitij Studio / क्षितिज स्टूडियो	384	3:7
Tarini Studio / तारिणी स्टूडियो	320	3:8
Dhruv Studio / ध्रुव स्टूडियो	384	5:2
Neelam Studio / नीलम स्टूडियो	152	4:5
Parth Studio / पार्थ स्टूडियो	304	2:1

Q1: Find the difference between the number of female members in Dhruv Studio and the number of male members in Kshitij Studio.

ध्रुव स्टूडियो में महिला सदस्यों और क्षितिज स्टूडियो में पुरुष सदस्यों की संख्या के बीच का अंतर ज्ञात कीजिए।

- (A) 40
- (B) 24
- (C) 32
- (D) 48
- (E) 56

Q2: If the ratio of total female members in Kshitij Studio, Neelam Studio and Parth Studio together to total male members in Tarini Studio, Dhruv Studio and Neelam Studio together is P:Q, then find P + Q.

यदि क्षितिज स्टूडियो, नीलम स्टूडियो और पार्थ स्टूडियो में मिलाकर कुल महिला सदस्यों तथा तारिणी स्टूडियो, ध्रुव स्टूडियो और नीलम स्टूडियो में मिलाकर कुल पुरुष सदस्यों का अनुपात $P:Q$ है, तो $P + Q$ ज्ञात कीजिए।

- (A) 367
- (B) 391
- (C) 383
- (D) 371
- (E) 379

Q3: In Manav Studio, the number of male members is double the average number of female members in Tarini Studio and Neelam Studio. If the number of female members in Manav Studio is three times the average number of male members in Kshitij Studio and Parth Studio, then find the total number of members in Manav Studio.

मानव स्टूडियो में पुरुष सदस्यों की संख्या, तारिणी स्टूडियो और नीलम स्टूडियो में महिला सदस्यों की औसत संख्या की दोगुनी है। यदि मानव स्टूडियो में महिला सदस्यों की संख्या, क्षितिज स्टूडियो और पार्थ स्टूडियो में पुरुष सदस्यों की औसत संख्या की तीन गुनी है, तो मानव स्टूडियो में कुल सदस्यों की संख्या ज्ञात कीजिए।

- (A) 2232
- (B) 2264
- (C) 2288
- (D) 2208
- (E) 2248

Q4: The number of female members in Parth Studio is what percentage more or less than the number of male members in Neelam Studio?

पार्थ स्टूडियो में महिला सदस्यों की संख्या, नीलम स्टूडियो में पुरुष सदस्यों की संख्या से कितने प्रतिशत अधिक या कम है?

- (A) 12.5%
- (B) 25%
- (C) 33.33%
- (D) 20%
- (E) 16.67%

Q5: Find the difference between the average number of male members in Kshitij Studio, Tarini Studio and Neelam Studio and the average number of female members in Tarini Studio, Dhruv Studio and Neelam Studio.

क्षितिज स्टूडियो, तारिणी स्टूडियो और नीलम स्टूडियो में पुरुष सदस्यों की औसत संख्या तथा तारिणी स्टूडियो, ध्रुव स्टूडियो और नीलम स्टूडियो में महिला सदस्यों की औसत संख्या के बीच का अंतर ज्ञात कीजिए।

- (A) 168
(B) 176
(C) 152
(D) 184
(E) 160

Solutions:

Studio / स्टूडियो	Female members / महिला सदस्य	Male members / पुरुष सदस्य	Total members / कुल सदस्य
Kshitij Studio / क्षितिज	288	672	960
Tarini Studio / तारिणी	192	512	704
Dhruv Studio / ध्रुव	640	256	896
Neelam Studio / नीलम	608	760	1368
Parth Studio / पार्थ	608	304	912

Common values:

Kshitij Studio: Female:Male = 3:7, difference = 4 parts = 384, so 1 part = 96

Female = 288, Male = 672

Tarini Studio: Female:Male = 3:8, difference = 5 parts = 320, so 1 part = 64

Female = 192, Male = 512

Dhruv Studio: Female:Male = 5:2, difference = 3 parts = 384, so 1 part = 128

Female = 640, Male = 256

Neelam Studio: Female:Male = 4:5, difference = 1 part = 152

Female = 608, Male = 760

Parth Studio: Female:Male = 2:1, difference = 1 part = 304

Female = 608, Male = 304

1. Option C

Female members in Dhruv Studio = 640

Male members in Kshitij Studio = 672

Required difference = 672 - 640 = 32

2. Option E

Female members in Kshitij, Neelam and Parth = $288 + 608 + 608 = 1504$

Male members in Tarini, Dhruv and Neelam = $512 + 256 + 760 = 1528$

Ratio = $1504:1528 = 188:191$

$P + Q = 188 + 191 = 379$

3. Option B

Male members in Manav Studio = $2 \times (192 + 608)/2 = 800$

Female members in Manav Studio = $3 \times (672 + 304)/2 = 1464$

Total members in Manav Studio = $800 + 1464 = 2264$

4. Option D

Female members in Parth Studio = 608

Male members in Neelam Studio = 760

Difference = $760 - 608 = 152$

Required percentage = $152/760 \times 100 = 20\%$ less

5. Option A

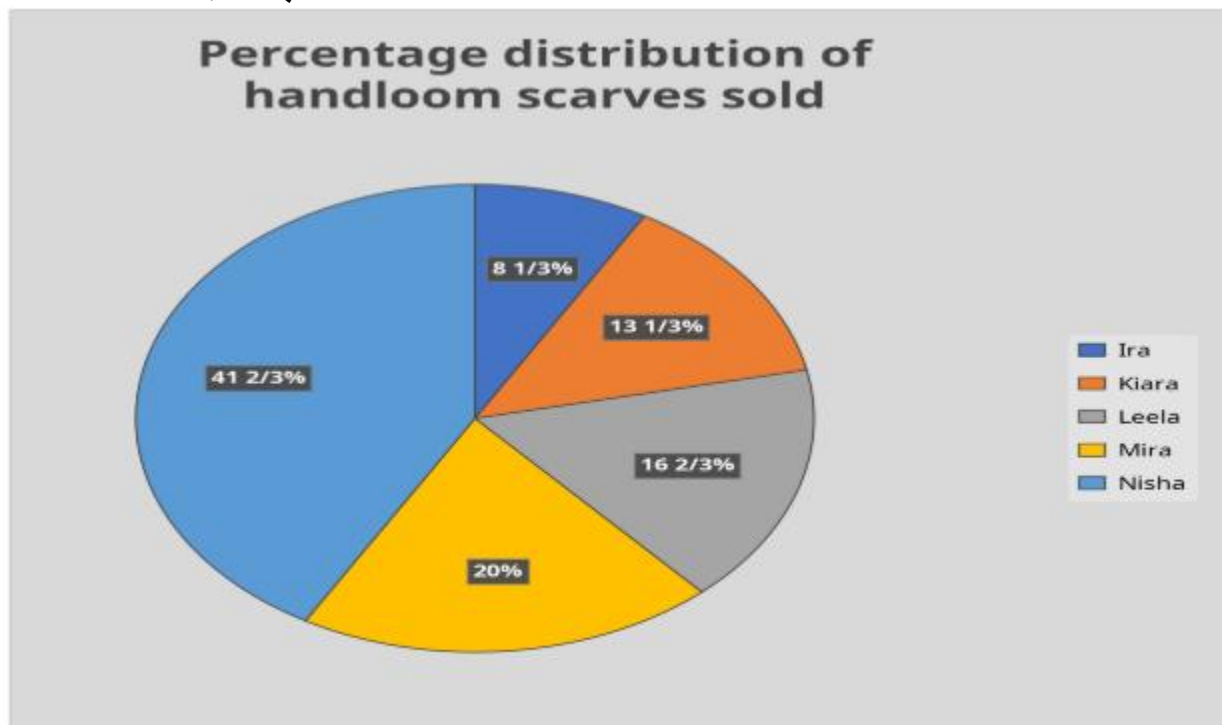
Average male members in Kshitij, Tarini and Neelam = $(672 + 512 + 760)/3 = 648$

Average female members in Tarini, Dhruv and Neelam = $(192 + 640 + 608)/3 = 480$

Required difference = $648 - 480 = 168$

Set 2: The pie chart shows the percentage distribution of the total number of handloom scarves sold by five boutiques: Ira, Kiara, Leela, Mira and Nisha. Read the pie chart carefully and answer the following questions.

पाई चार्ट पाँच बुटीक, इरा, कियारा, लीला, मीरा और निशा द्वारा बेचे गए कुल हैंडलूम स्कार्फ का प्रतिशत वितरण दर्शाता है। पाई चार्ट को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: If the average number of scarves sold by Leela, Mira and Nisha is 18800, then find the total number of scarves sold by all five boutiques together.

यदि लीला, मीरा और निशा द्वारा बेचे गए स्कार्फ की औसत संख्या 18800 है, तो पाँचों बुटीक द्वारा मिलाकर बेचे गए कुल स्कार्फ की संख्या ज्ञात कीजिए।

- (A) 64000
- (B) 68000
- (C) 76000
- (D) 72000
- (E) 80000

Q2: Total scarves sold by Ira and Kiara together are what percentage of total scarves sold by Leela?

इरा और कियारा द्वारा मिलाकर बेचे गए कुल स्कार्फ, लीला द्वारा बेचे गए कुल स्कार्फ का कितना प्रतिशत हैं?

- (A) 130%

- (B) 125%
- (C) 140%
- (D) 120%
- (E) 150%

Q3: Find the ratio of total scarves sold by Mira and Nisha together to total scarves sold by Ira and Kiara together.

मीरा और निशा द्वारा मिलाकर बेचे गए कुल स्कार्फ का इरा और कियारा द्वारा मिलाकर बेचे गए कुल स्कार्फ से अनुपात ज्ञात कीजिए।

- (A) 31:13
- (B) 35:17
- (C) 39:11
- (D) 17:37
- (E) 37:13

Q4: 25% of the scarves sold by Kiara are returned by customers and 12.5% of the scarves sold by Leela are returned by customers. If the total number of scarves returned by customers from Kiara and Leela together is 3900, then find the number of scarves sold by Nisha.

कियारा द्वारा बेचे गए 25% स्कार्फ ग्राहकों द्वारा वापस कर दिए गए और लीला द्वारा बेचे गए 12.5% स्कार्फ ग्राहकों द्वारा वापस कर दिए गए। यदि कियारा और लीला से मिलाकर ग्राहकों द्वारा वापस किए गए स्कार्फ की कुल संख्या 3900 है, तो निशा द्वारा बेचे गए स्कार्फ की संख्या ज्ञात कीजिए।

- (A) 28000
- (B) 30000
- (C) 32000
- (D) 36000
- (E) 24000

Q5: Find the central angle for total scarves sold by Ira and Kiara together.

इरा और कियारा द्वारा मिलाकर बेचे गए कुल स्कार्फ के लिए केंद्रीय कोण ज्ञात कीजिए।

- (A) 72°
- (B) 84°
- (C) 78°
- (D) 90°
- (E) 96°

Solutions:

Boutique	Percentage distribution	Central angle
Ira	$8\frac{1}{3}\%$	30°
Kiara	$13\frac{1}{3}\%$	48°
Leela	$16\frac{2}{3}\%$	60°
Mira	20%	72°
Nisha	$41\frac{2}{3}\%$	150°

1. Option D

$$\text{Leela} + \text{Mira} + \text{Nisha} = 16\frac{2}{3}\% + 20\% + 41\frac{2}{3}\% = 78\frac{1}{3}\% = \frac{47}{60}$$

$$\text{Total scarves sold by Leela, Mira and Nisha} = 18800 \times 3 = 56400$$

$$\frac{47}{60} \text{ of total} = 56400$$

$$\text{Total scarves sold} = 56400 \times \frac{60}{47} = 72000$$

2. Option A

$$\text{Ira} + \text{Kiara} = 8\frac{1}{3}\% + 13\frac{1}{3}\% = 21\frac{2}{3}\% = \frac{13}{60}$$

$$\text{Leela} = 16\frac{2}{3}\% = \frac{10}{60}$$

$$\text{Required percentage} = \frac{13}{10} \times 100 = 130\%$$

3. Option E

$$\text{Mira} + \text{Nisha} = 20\% + 41\frac{2}{3}\% = 61\frac{2}{3}\% = \frac{37}{60}$$

$$\text{Ira} + \text{Kiara} = 8\frac{1}{3}\% + 13\frac{1}{3}\% = 21\frac{2}{3}\% = \frac{13}{60}$$

$$\text{Required ratio} = \frac{37}{60} : \frac{13}{60} = 37:13$$

4. Option B

$$\text{Let total scarves sold} = T$$

$$\text{Kiara returned} = 25\% \text{ of } 13\frac{1}{3}\% \text{ of } T = \frac{1}{4} \times \frac{8}{60}T = \frac{2}{60}T$$

$$\text{Leela returned} = 12.5\% \text{ of } 16\frac{2}{3}\% \text{ of } T = \frac{1}{8} \times \frac{10}{60}T = \frac{10}{480}T$$

$$\text{Total returned} = \frac{2}{60}T + \frac{10}{480}T = \frac{13}{240}T$$

$$\frac{13}{240}T = 3900$$

$$T = 72000$$

$$\text{Nisha scarves sold} = 41\frac{2}{3}\% \text{ of } 72000 = \frac{5}{12} \times 72000 = 30000$$

5. Option C

Ira central angle = 30°

Kiara central angle = 48°

Required central angle = $30^\circ + 48^\circ = 78^\circ$

CHECKLIST

BY

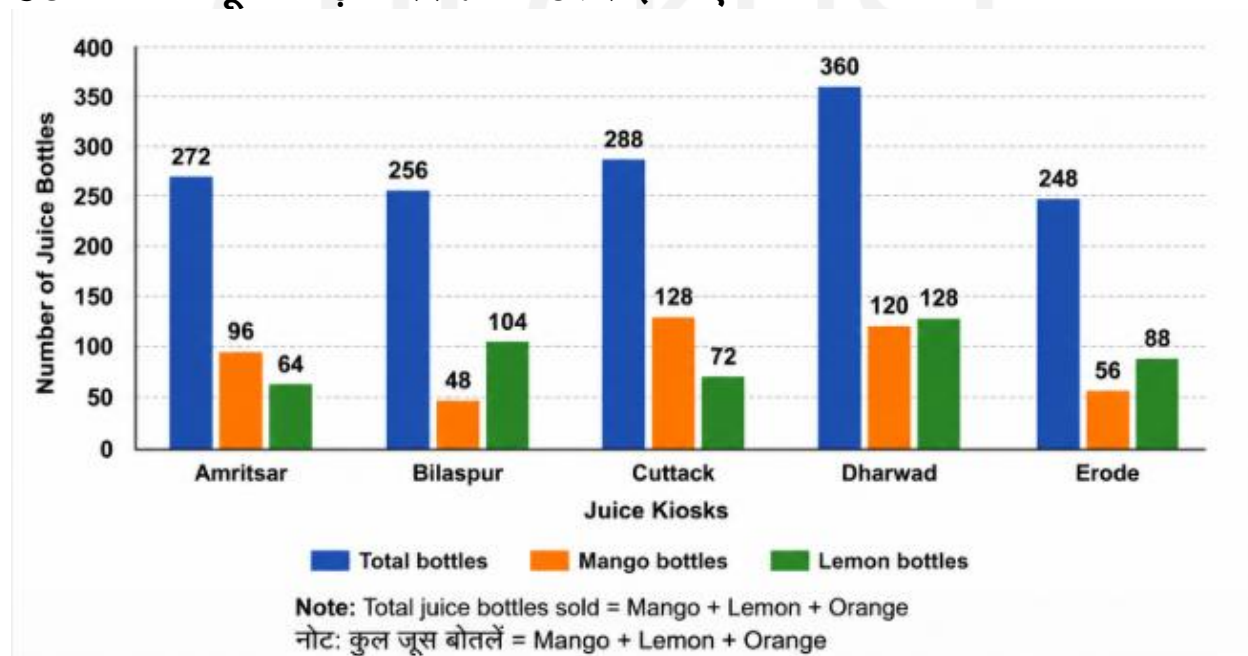
AASHISH

ARORA

Set 3: The bar graph shows the total number of juice bottles sold by five different juice kiosks and the number of Mango and Lemon juice bottles sold by these kiosks. Only Mango, Lemon and Orange juice bottles are sold by these kiosks. Orange juice bottles = Total juice bottles – Mango juice bottles – Lemon juice bottles. Read the data carefully and answer the questions.

दंड आलेख पाँच अलग-अलग जूस कियोस्क द्वारा बेची गई कुल जूस बोतलों और इन कियोस्क द्वारा बेची गई Mango तथा Lemon जूस बोतलों की संख्या दर्शाता है। इन कियोस्क पर केवल Mango, Lemon और Orange जूस बोतलें ही बेची जाती हैं।

Orange जूस बोतलें = कुल जूस बोतलें – Mango जूस बोतलें – Lemon जूस बोतलें।
डेटा को ध्यानपूर्वक पढ़ें और प्रश्नों के उत्तर दीजिए।



Q1: Find the maximum number of Orange juice bottles sold by any one juice kiosk.

किसी एक जूस कियोस्क द्वारा बेची गई Orange जूस बोतलों की अधिकतम संख्या ज्ञात कीजिए।

- (A) 96
- (B) 104
- (C) 120
- (D) 128
- (E) 112

Q2: Find the average number of Orange juice bottles sold by all five juice kiosks.

सभी पाँच जूस कियोस्क द्वारा बेची गई Orange जूस बोतलों की औसत संख्या ज्ञात कीजिए।

- (A) 108

- (B) 104
- (C) 112
- (D) 116
- (E) 100

Q3: Find the ratio of Mango juice bottles sold by Amritsar and Cuttack together to Orange juice bottles sold by Dharwad and Erode together. If the required ratio is $a:b$, find $a + b$.

अमृतसर और कटक द्वारा मिलाकर बेची गई Mango जूस बोतलों का धारवाड़ और इरोड द्वारा मिलाकर बेची गई Orange जूस बोतलों से अनुपात ज्ञात कीजिए। यदि आवश्यक अनुपात $a:b$ है, तो $a + b$ ज्ञात कीजिए।

- (A) 49
- (B) 51
- (C) 53
- (D) 55
- (E) 57

Q4: If the total juice bottles sold by a new kiosk Nandan are 20% more than the total juice bottles sold by Dharwad, and the ratio of Mango, Lemon and Orange juice bottles sold by Nandan is 5:6:7, then find the number of Orange juice bottles sold by Nandan.

यदि एक नए कियोस्क नंदन द्वारा बेची गई कुल जूस बोतलें, धारवाड़ द्वारा बेची गई कुल जूस बोतलों से 20% अधिक हैं, और नंदन द्वारा बेची गई Mango, Lemon और Orange जूस बोतलों का अनुपात 5:6:7 है, तो नंदन द्वारा बेची गई Orange जूस बोतलों की संख्या ज्ञात कीजिए।

- (A) 168
- (B) 160
- (C) 176
- (D) 184
- (E) 192

Q5: Total Orange juice bottles sold by Cuttack, Dharwad and Erode together are how many more or less than total Mango juice bottles sold by Amritsar, Bilaspur and Cuttack together?

कटक, धारवाड़ और इरोड द्वारा मिलाकर बेची गई कुल Orange जूस बोतलें, अमृतसर, बिलासपुर और कटक द्वारा मिलाकर बेची गई कुल Mango जूस बोतलों से कितनी अधिक या कम हैं?

- (A) 24
(B) 28
(C) 32
(D) 36
(E) 40

Solutions:

Juice Kiosk / जूस कियोस्क	Total Bottles / कुल बोतलें	Mango / Mango	Lemon / Lemon	Orange Bottles / Orange बोतलें
Amritsar	272	96	64	$272 - (96 + 64) = 112$
Bilaspur	256	48	104	$256 - (48 + 104) = 104$
Cuttack	288	128	72	$288 - (128 + 72) = 88$
Dharwad	360	120	128	$360 - (120 + 128) = 112$
Erode	248	56	88	$248 - (56 + 88) = 104$

1. Option E

Orange in Amritsar = $272 - 96 - 64 = 112$
 Orange in Bilaspur = $256 - 48 - 104 = 104$
 Orange in Cuttack = $288 - 128 - 72 = 88$
 Orange in Dharwad = $360 - 120 - 128 = 112$
 Orange in Erode = $248 - 56 - 88 = 104$
 Maximum Orange juice bottles = 112

2. Option B

Orange bottles = 112, 104, 88, 112 and 104
 Average = $(112 + 104 + 88 + 112 + 104) / 5$
 Average = $520 / 5 = 104$

3. Option D

Mango in Amritsar and Cuttack together = $96 + 128 = 224$
 Orange in Dharwad = $360 - 120 - 128 = 112$
 Orange in Erode = $248 - 56 - 88 = 104$
 Orange in Dharwad and Erode together = $112 + 104 = 216$
 Required ratio = $224:216 = 28:27$

$$a + b = 28 + 27 = 55$$

4. Option A

Total bottles sold by Dharwad = 360

Total bottles sold by Nandan = $360 \times \frac{6}{5} = 432$

Ratio of Mango:Lemon:Orange = 5:6:7

Total parts = 18

Orange bottles sold by Nandan = $432 \times \frac{7}{18} = 168$

5. Option C

Orange in Cuttack = $288 - 128 - 72 = 88$

Orange in Dharwad = $360 - 120 - 128 = 112$

Orange in Erode = $248 - 56 - 88 = 104$

Total Orange in Cuttack, Dharwad and Erode = $88 + 112 + 104 = 304$

Total Mango in Amritsar, Bilaspur and Cuttack = $96 + 48 + 128 = 272$

Difference = $304 - 272 = 32$ more

Set 4: A hobby academy conducted a survey among 1200 students to study their enrollment in three hobby classes: Tabla, Kathak and Madhubani Painting. Each student is enrolled in at least one of these three classes. The ratio of students enrolled only in Tabla, only in Kathak and only in Madhubani Painting is 11 : 10 : 7, respectively. The ratio of students enrolled only in Tabla and Kathak, only in Kathak and Madhubani Painting and only in Tabla and Madhubani Painting is 13 : 17 : 27, respectively. The number of students enrolled in all three hobby classes is 72. The number of students enrolled in more than one hobby class is 44% of the total number of students surveyed.

एक हॉबी अकादमी ने 1200 विद्यार्थियों पर एक सर्वे किया ताकि तीन हॉबी क्लास: तबला, कथक और मधुबनी पेंटिंग में उनके नामांकन का अध्ययन किया जा सके। प्रत्येक विद्यार्थी इन तीनों में से कम से कम एक क्लास में नामांकित है। केवल तबला, केवल कथक और केवल मधुबनी पेंटिंग में नामांकित विद्यार्थियों का अनुपात क्रमशः 11 : 10 : 7 है। केवल तबला और कथक, केवल कथक और मधुबनी पेंटिंग तथा केवल तबला और मधुबनी पेंटिंग में नामांकित विद्यार्थियों का अनुपात क्रमशः 13 : 17 : 27 है। तीनों हॉबी क्लास में नामांकित विद्यार्थियों की संख्या 72 है। एक से अधिक हॉबी क्लास में नामांकित विद्यार्थियों की संख्या, सर्वे किए गए कुल विद्यार्थियों की संख्या का 44% है।

Q1: Find the number of students who are enrolled in Madhubani Painting.

मधुबनी पेंटिंग में नामांकित विद्यार्थियों की संख्या ज्ञात कीजिए।

- (A) 568
- (B) 580
- (C) 604
- (D) 592
- (E) 616

Q2: Find the number of students enrolled in at most two hobby classes as a percentage of the total number of students surveyed.

अधिकतम दो हॉबी क्लास में नामांकित विद्यार्थियों की संख्या, सर्वे किए गए कुल विद्यार्थियों का कितना प्रतिशत है?

- (A) 94%
- (B) 90%
- (C) 96%
- (D) 88%
- (E) 92%

Q3: The number of students enrolled in only one hobby class is how much more than the number of students enrolled in more than one hobby class?

केवल एक हॉबी क्लास में नामांकित विद्यार्थियों की संख्या, एक से अधिक हॉबी क्लास में नामांकित विद्यार्थियों की संख्या से कितनी अधिक है?

- (A) 128
- (B) 168
- (C) 152
- (D) 136
- (E) 144

Q4: Find the ratio of the number of students enrolled only in Tabla and only in Kathak together to the number of students enrolled only in Madhubani Painting and only in Kathak together.

केवल तबला और केवल कथक में नामांकित विद्यार्थियों की कुल संख्या का, केवल मधुबनी पेंटिंग और केवल कथक में नामांकित विद्यार्थियों की कुल संख्या से अनुपात ज्ञात कीजिए।

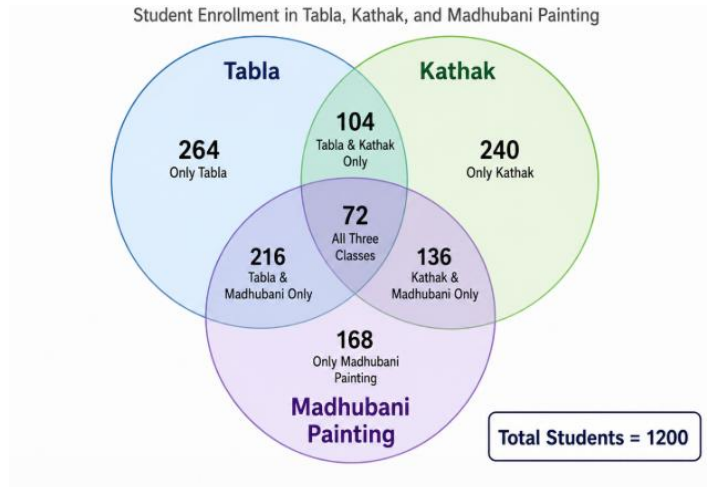
- (A) 17 : 21
- (B) 15 : 14
- (C) 21 : 17
- (D) 11 : 9
- (E) 19 : 16

Q5: If the ratio of boys to girls among all surveyed students is 2 : 3 and the ratio of boys to girls among students enrolled in all three hobby classes is 2 : 1, then the number of boys enrolled in all three hobby classes is what percentage of the total number of boys surveyed?

यदि सर्वे किए गए सभी विद्यार्थियों में लड़कों और लड़कियों का अनुपात 2 : 3 है और तीनों हॉबी क्लास में नामांकित विद्यार्थियों में लड़कों और लड़कियों का अनुपात 2 : 1 है, तो तीनों हॉबी क्लास में नामांकित लड़कों की संख्या, सर्वे किए गए कुल लड़कों की संख्या का कितना प्रतिशत है?

- (A) 9%
- (B) 10%
- (C) 12%
- (D) 8%
- (E) 15%

Solutions:



1. Option D

Students enrolled in Madhubani Painting = Only Madhubani Painting + Only Kathak and Madhubani Painting + Only Tabla and Madhubani Painting + All three

$$= 168 + 136 + 216 + 72$$

$$= 592$$

2. Option A

Students enrolled in all three hobby classes = 72

Students enrolled in at most two hobby classes = $1200 - 72$

$$= 1128$$

Required percentage = $1128 \div 1200 \times 100$

$$= 94\%$$

3. Option E

Students enrolled in more than one hobby class = 44% of 1200

$$= 528$$

Students enrolled in only one hobby class = $1200 - 528$

$$= 672$$

Required difference = $672 - 528$

$$= 144$$

4. Option C

Only Tabla + Only Kathak = $264 + 240$

$$= 504$$

Only Madhubani Painting + Only Kathak = $168 + 240$

$$= 408$$

Required ratio = 504 : 408
= 21 : 17

5. Option B

Total boys surveyed = $\frac{2}{5}$ of 1200
= 480

Boys enrolled in all three hobby classes = $\frac{2}{3}$ of 72
= 48

Required percentage = $48 \div 480 \times 100$
= 10%

CHECKLIST
BY
AASHISH
ARORA